

SSN College of Engineering, Kalavakkam
Department of Computer Science and Engineering
V Semester - CSE 'A'
UCS2511 NETWORKS LAB
Exercise 12
TCP UDP Performance Evaluation

Name : Aathish Sivasubrahmanian 3122215001001

Aim:

Simulation of TCP and UDP protocols and their comparison.

Code

```
#Create a simulator object set ns [new Simulator] #Open the nam trace file set nf
[open out.nam w] $ns namtrace-all $nf

$ns color 1 Blue
$ns color 2 Red
#Define a 'finish' procedure proc finish {} {

global ns nf
$ns flush-trace
#Close the trace file
close $nf
#Execute nam on the trace file exec nam out.nam &
exit 0

}
# Creating Nodes
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
```

```

set n4 [$ns node]
set n5 [$ns node]
#Setting Links
$ns duplex-link $n0 $n2 2Mb 10ms DropTail $ns duplex-link $n1 $n2 2Mb 10ms
DropTail $ns duplex-link $n2 $n3 0.3Mb 100ms DropTail $ns duplex-link $n3 $n2
0.3Mb 100ms DropTail $ns duplex-link $n3 $n4 0.5Mb 40ms DropTail $ns
duplex-link $n3 $n5 0.5Mb 40ms DropTail #Setting Topology
$ns duplex-link-op $n0 $n2 orient right-down $ns duplex-link-op $n1 $n2 orient
right-up
$ns duplex-link-op $n2 $n3 orient left-up

$ns duplex-link-op $n3 $n2 orient left
$ns duplex-link-op $n3 $n4 orient up
$ns duplex-link-op $n3 $n5 orient right-up
#Setting Queue Limit
$ns queue-limit $n2 $n3 10
#Setup a TCP connection over 0 and 4 and its flow id, window size, packet size
set tcp [new Agent/TCP/Newreno]
$ns attach-agent $n0 $tcp
set sink [new Agent/TCPSink/DelAck]
$ns attach-agent $n4 $sink
$ns connect $tcp $sink
$tcp set fid_ 1
$tcp set window_ 8000
$tcp set packetSize_ 552
#Setup a FTP over TCP connection
set ftp [new Application/FTP]
$ftp attach-agent $tcp
$ftp set type_ FTP
#Create a UDP agent and attach it to node n0
set udp [new Agent/UDP]
$ns attach-agent $n0 $udp
# Create a CBR traffic source and attach it to udp0
set cbr [new Application/Traffic/CBR]
$cbr set type_ CBR
$cbr set packet_size_ 1000
$cbr set rate_ 0.01mb
$cbr set random_ false
$cbr attach-agent $udp

```

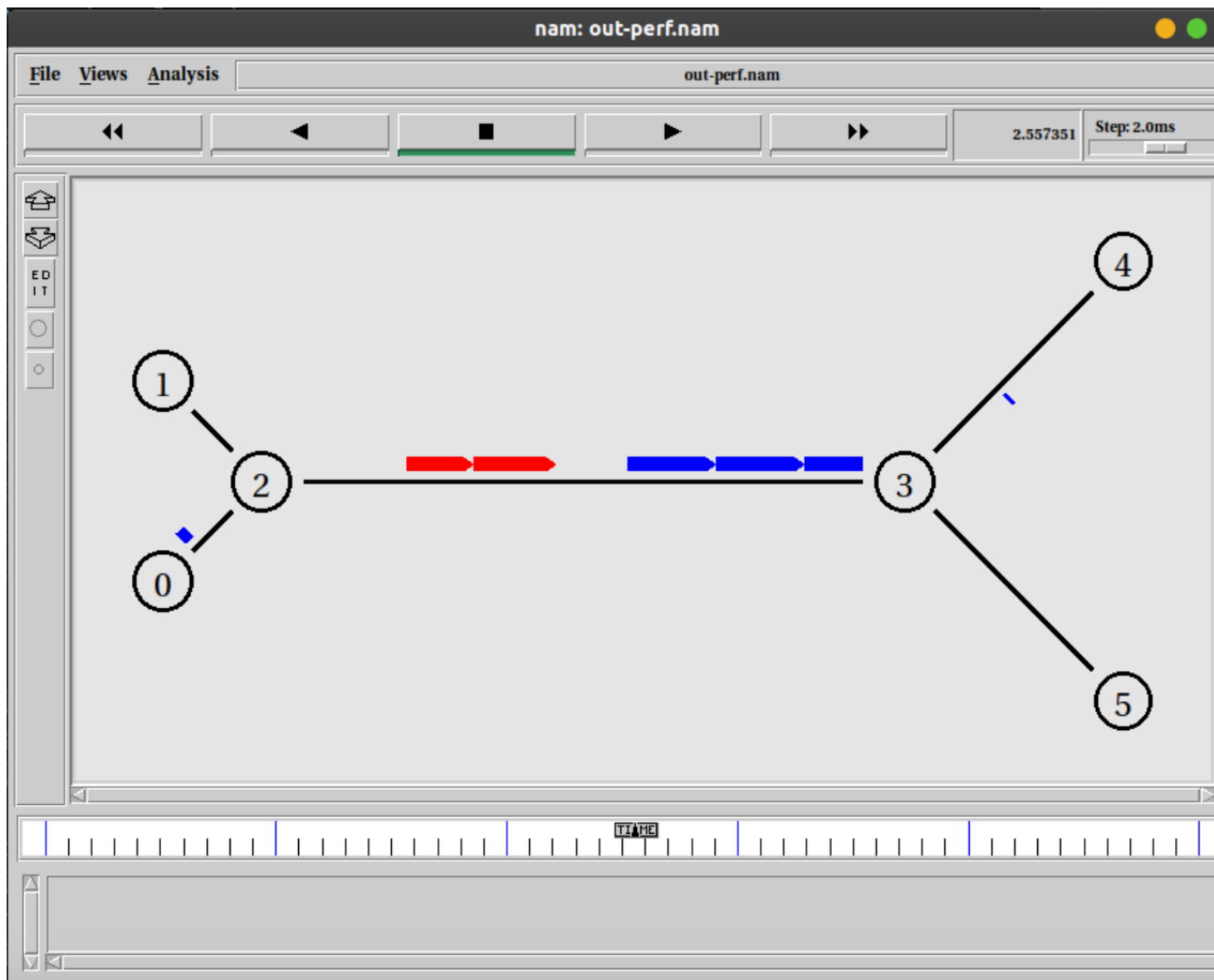
```
#Create a Null agent (a traffic sink) and attach it to node n1 set null [new
Agent/Null]
$ns attach-agent $n5 $null
#Connect the traffic source with the traffic sink
$ns connect $udp $null
#Set Flow ID, Packet Size and Window Size
$udp set fid_ 2
$udp set window_ 8000
$udp set packetSize_ 552
#Start and stop the cbr and ftp

$ns at 0.1 "$cbr start"
$ns at 1.0 "$ftp start"
$ns at 4.5 "$ftp stop"
$ns at 5.0 "$cbr stop"
#Call the finish procedure after 5 seconds of simulation time $ns at 5.0 "finish"

#Run the simulation

$ns run
```

Output:



Learning Outcomes:

I learned how to compare TCP and UDP protocols.